Emotional Responses

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Living With Hypoglycemia

This example is extended from Living With Hypoglycemia: An Exploration of Patients' Emotions: Qualitative Findings From the InHypo-DM Study, Canada2019 publication.

[The purpose of this code is to examine how well LLMs can recover emotional states that were human coded in the publication without information about themes or context other than a user prompt. We use Claude Sonnet 3.7, an LLM optimized for computational efficiency and reliability in research applications, offering robust performance for systematic analysis, data processing, and academic workflows through multiple access including the use of an API. Anthropic, creator of Claude, takes user security and privacy seriously and conversations are not used for model training such as in other LLM platforms.]The purpose of this code is to examine how well LLMs can recover emotional states that were human coded in the publication without information about themes or context other than a user prompt. We use Claude Sonnet 3.7, an LLM optimized for computational efficiency and reliability in research applications, offering robust performance for systematic analysis, data processing, and academic workflows through multiple access including the use of an API.[1] Anthropic, creator of Claude, takes user security and privacy seriously and conversations are not used for model training such as in other LLM platforms.

Setting up Claude Sonnet 3.7 is simple via rstudio using the elmer package but requires an API key and a credit card on file.

[1]Submitting an API request costs $.05 and can use up to 1 million tokens. A token is a basic unit of text that AI language models process, typically representing words, parts of words, or punctuation marks.

```r
library(ellmer)
library(tidyverse)

# Note.
## initialize connection after adding api key to .Renvironment file
## usethis::edit_r_environ() opens file
```

In using LLMs it is important to provide it with a robust prompt that contains a single task and set of rules. Below we generate a simple prompt with basic rules. It is possible to provide the LLM a pdf document with a systematic coding schema.

```r
# test 2 with data
system_prompt <- paste(
  # task
  "You are a graduate researcher that will identify and categorize
  emotional states expressed by participants during interviews about
  their adversities living with type 1 and type 2 diabetes to better
  understand hypoglycemia.",
  # rules
  "#keep the response brief",
  "* Use examples when helpful.",
  # format
  collapse = "\n"
)
```

We are also interested in how a local (e.g., smaller and less robust model) LLM such as llama 3.2 which is installed locally via ollama, performs on this task. The advantage of using llama is that it is installed locally which translates as a secured way of processing data and it is free as it uses local computing resources. The disadvantage is that these models tend to be much smaller compared to Claude, ChatGPT, and Gemini.

We compare both models using the same user prompt. First we print the text from the paper, followed by the Claude, and llama assessments before moving on to the next text example.

```r
client_claude <- chat_anthropic(system_prompt,
                                model = "claude-3-7-sonnet-latest")

client_llama <- chat_ollama(system_prompt,
                            model = "llama3.2:3b")
```

Finally, we pass it examples from interviews found in the public on page 272, results section.

The analysis revealed an intricate pattern describing the participants' emotions regarding their experiences of hypoglycemia, including fear, anxiety, frustration, confidence, and hope, and how these emotions initiated a call to action. A predominant emotion expressed by participants was fear.

> *"Usually going low at night because it scares me, and I think that is why I can't sleep very well, because I am always scared at nighttime that I will drop into a low and I won't feel it, because I have had that happen to me once before where I woke up in the hospital. So, I think that is the main concern I have.... I get scared. I get really scared.* (T1F #1)

```
string_ex <- "Usually going low at night because it scares me,
and I think that is why I can’t sleep very well, because I
am always scared at nighttime that I will drop into a low
and I won’t feel it, because I have had that happen to me
once before where I woke up in the hospital. So, I think
that is the main concern I have…. I get scared. I get
really scared."

client_claude$chat(string_ex)
```

```
The participant is expressing fear and anxiety, particularly about nighttime
hypoglycemia. Their emotional state includes:
```

```
1. Fear ("it scares me," "I am always scared," "I get scared. I get really
scared")
2. Anxiety about sleeping due to worry about undetected low blood sugar
3. Trauma response related to a previous severe hypoglycemic event that
resulted in hospitalization

This appears to be a manifestation of hypoglycemia-associated anxiety,
specifically nocturnal hypoglycemia fear that is reinforced by past traumatic
experience.
```

```r
client_llama$chat(string_ex)
```

```
This participant is expressing:

* Anxiety: "scared" and "really scared"
* Fear of hypoglycemia: specifically night-time lows that can be
misattributable (not feeling it before waking up in the hospital)
* Distress: experiencing poor sleep quality due to fear
* Concern: having had a serious experience with a low that was not caught,
leading to hospitalization

Categorization: Emotional state - Anxiety with a Fear component
```

There was a clear link between fear and anxiety. Participants described how, as their blood glucose began to drop at an unpredictable rate, their worry increased, as did their subsequent anxiety about the best action to remedy the situation. For participants who were living alone, this combination of fear and anxiety was intensified.

> *"It's when it [blood glucose] is under…usually under 5 [mmol/L] I start getting worried and scared that I don't know how fast it's going to drop. I don't know if I will get there [remedy for hypoglycemia] on time, especially when I am by myself at home."* (T1F #1)

```
string_ex <- "It’s when it [blood glucose] is under…usually
under 5 [mmol/L] I start getting worried and scared that
I don’t know how fast it’s going to drop. I don’t
know if I will get there [remedy for hypoglycemia] on
time, especially when I am by myself at home."

client_claude$chat(string_ex)
```

```
The participant is expressing:

1. Anxiety/worry ("I start getting worried")
2. Fear ("scared") related to:
   - Unpredictability of blood glucose drops
```

```
   - Uncertainty about treatment timing
   - Vulnerability when alone

This represents hypoglycemia-related anxiety with specific concerns about:
- The rate of blood glucose decline
- Access to treatment
- Isolation as an exacerbating factor for their fear

Their emotional response is triggered at specific blood glucose thresholds
(under 5 mmol/L), showing how biomedical measurements directly influence
psychological states.
```

```r
client_llama$chat(string_ex)
```

```
This participant is expressing:

* Fear of speed: concerned about the severity and speed of a hypoglycemic
episode
* Powerlessness: feeling unsure if they can respond in time to prevent harm
(difficulty getting remedy on time while alone)
* Anxiety related to self-management: worried that without assistance, they may
not be able to recover from an episode

Categorization: Emotional state - Anxiety with a Fear of Loss of Control
component
```

A sense of hope in the capacity to control their hypoglycemia events was rooted in both their own ability to manage their hypoglycemia and in outside sources of confidence, such as education.

> *"Just the fact that everyone can learn stuff and do things differently all the time, so even if there was something that I was doing wrong, if you learn a new behavior, then it changes what you're doing, so that makes me hopeful."* (T1F #5)

```r
string_ex <- "Just the fact that everyone can learn stuff
and do things differently all the time, so even if there
was something that I was doing wrong, if you learn a new
behavior, then it changes what you’re doing, so that
makes me hopeful."

client_claude$chat(string_ex)
```

```
The participant is expressing:

1. Hope ("makes me hopeful") - the primary emotional state
2. Optimism about the possibility of positive change
3. Growth mindset - belief that learning new behaviors can lead to improvement

This reflects resilience and a positive outlook regarding their diabetes
management. The participant finds emotional comfort in the idea that
adaptability and learning can lead to better outcomes, demonstrating
psychological coping through maintaining hope for improvement despite
challenges with diabetes management.
```

```r
client_llama$chat(string_ex)
```

```
This participant is expressing:

* Optimism: feeling hopeful about being able to change and find ways to manage
hypoglycemia
* Acceptance of variability: acknowledging that everyone has different
approaches and experiences
* Hope for resilience: learning can help mitigate the impact of past mistakes
or difficulties

Categorization: Emotional state - Optimism with a Resilience component
```

These examples clearly show the advantage of using LLMs for this research as they come very close to human coding with minimal instructions. We do not recommend a full replacement of human coders but rather to integrate LLMs into a coding schema workflow in order to compliment data analysis. Claude tended to perform slightly better than llama using these examples. However, these examples from the paper were probably chosen for their clear indications of the emotional states of interest to the researchers. We are uncertain how these LLMs would perform with ambiguous examples of emotional states at this point.